413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 10

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In the previous lecture, we studied different approaches for estimating unknown parameters from observed data. Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} p_\theta(x),$$

where θ is an unknown parameter we would like to estimate.

Method of moments. The method of moments estimates θ by matching population moments to their empirical counterparts:

$$\mathbb{E}_\theta[X^k] \quad \longleftrightarrow \quad \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

Sampling variability. Since an estimator depends on the random sample,

$$\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n),$$

it is itself a random variable. Its standard error describes how much the estimate would vary if the experiment were repeated:

$$SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}.$$

When the sampling distribution is difficult to derive analytically, bootstrap methods can be used to approximate it numerically.

Maximum likelihood estimation. The likelihood of the observed data $x_1, \ldots, x_n$ is

$$L(\theta) = \prod_{i=1}^{n} p_\theta(x_i).$$

The maximum likelihood estimator is

$$\hat{\theta}_{\text{MLE}} = \arg\max_\theta L(\theta) = \arg\max_\theta \ell(\theta),$$

The natural next question is:

How accurate is $\hat{\theta}_{\text{MLE}}$ as an estimate of the true parameter?

1 Large-sample properties of maximum likelihood estimators

In the previous lecture, we defined the maximum likelihood estimator as

$$\hat{\theta}_{\text{MLE}} = \arg\max_\theta \ell(\theta),$$

where, for i.i.d. observations,

$$\ell(\theta) = \sum_{i=1}^{n} \log p_\theta(X_i).$$

Since $\hat{\theta}_{\text{MLE}}$ depends on the random sample $X_1, \ldots, X_n$, it is itself a random variable. We therefore want to understand its sampling variability.

1.1 Consistency

Suppose

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} p_{\theta_0}(x),$$

where θ_0 denotes the true parameter. Under suitable regularity conditions, the maximum likelihood estimator is consistent:

$$\boxed{\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta_0 \qquad \text{as } n \to \infty.}$$

Thus, as the sample size increases, the estimator becomes increasingly concentrated around the true parameter. Consistency should not be confused with unbiasedness. An estimator may be biased for finite n and still converge to the true parameter as $n \to \infty$.

1.2 Asymptotic normality

For sufficiently large n, the sampling distribution of the MLE is approximately Gaussian:

$$\hat{\theta}_{\text{MLE}} \approx \mathcal{N}\left(\theta_0, \frac{1}{nI(\theta_0)}\right).$$

Equivalently,

$$\sqrt{n}\left(\hat{\theta}_{\text{MLE}} - \theta_0\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right).$$

where $I(\theta)$ is the Fisher information

$$I(\theta) = \mathbb{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\log p_\theta(X)\right)^2\right].$$

1.3 Cramér–Rao lower bound

For any unbiased estimator $\hat{\theta}$ satisfying suitable regularity conditions,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}.$$

This is the Cramér–Rao lower bound. It gives a fundamental lower bound on the variance of an unbiased estimator. For large n, the MLE satisfies

$$\text{Var}\left(\hat{\theta}_{\text{MLE}}\right) \approx \frac{1}{nI(\theta_0)}.$$

Therefore, the maximum likelihood estimator asymptotically attains the Cramér–Rao lower bound and is said to be **asymptotically efficient**. In summary,

$$\hat{\theta}_{\text{MLE}} \approx \mathcal{N}\left(\theta_0, \frac{1}{nI(\theta_0)}\right)$$

for sufficiently large n. Thus, the uncertainty in the MLE decreases at the usual $1/\sqrt{n}$ rate as more data are collected.

2 From Maximum Likelihood to Bayesian Inference

So far, our goal in parameter estimation has been to use observed data

$$x_1, \ldots, x_n$$

to estimate an unknown parameter θ of a probabilistic model

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} p_\theta(x).$$

We first considered the **method of moments**, which estimates θ by matching population moments to their empirical counterparts. We then introduced **maximum likelihood estimation**, which uses the full probabilistic model and chooses the parameter value under which the observed dataset is most likely:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} p(\mathbf{x} \mid \theta).$$

Equivalently,

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \log p(\mathbf{x} \mid \theta).$$

Both methods produce a single estimate of the unknown parameter:

$$\mathbf{x} \longrightarrow \hat{\theta}.$$

A natural question is whether the observed data should be the *only* source of information about θ.

2.1 Incorporating prior information: MAP estimation

Maximum likelihood can be sensitive to the particular sample that was observed, especially when the sample size is small. For example, suppose

$$X_i \sim \text{Bernoulli}(\theta),$$

where θ is the probability of observing heads. If

$$X_1 = X_2 = X_3 = 1,$$

then

$$\hat{\theta}_{\text{MLE}} = 1.$$

Thus, based only on these three observations, the MLE assigns the largest likelihood to a coin that always lands heads. However, we may already have some information about plausible values of θ. For example, before observing the three flips, we may believe that an ordinary coin is likely to have θ reasonably close to $1/2$. We can represent such information through a probability distribution on the parameter,

$$p(\theta),$$

called the **prior distribution**. Using Bayes' rule,

$$\boxed{p(\theta \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \theta)p(\theta)}{p(\mathbf{x})}.}$$

Here,

$$\underbrace{p(\mathbf{x} \mid \theta)}_{\text{likelihood}}, \qquad \underbrace{p(\theta)}_{\text{prior}}, \qquad \underbrace{p(\theta \mid \mathbf{x})}_{\text{posterior}}.$$

If we still want a single point estimate of θ, we can choose the value that maximizes the posterior distribution directly:

$$\boxed{\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid \mathbf{x}).}$$

Since $p(\mathbf{x})$ does not depend on θ,

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\mathbf{x} \mid \theta)p(\theta).$$

Taking logarithms gives

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \left[\log p(\mathbf{x} \mid \theta) + \log p(\theta)\right].$$

Thus, MAP estimation extends maximum likelihood by combining information from the observed data with prior information about the parameter:

$$\boxed{\text{MAP} = \text{likelihood information} + \text{prior information}.}$$

If the prior is approximately constant over the relevant range of θ, then $\log p(\theta)$ contributes only an approximately constant term, and

$$\hat{\theta}_{\text{MAP}} \approx \hat{\theta}_{\text{MLE}}.$$

In this sense, maximum likelihood can be viewed as a limiting case of MAP estimation when the prior contributes very little information.

2.2 Beyond a point estimate: Bayesian inference

MAP estimation still reduces everything we know about θ to a single number,

$$\hat{\theta}_{\text{MAP}}.$$

But Bayes' rule gives us something richer: the entire posterior distribution

$$p(\theta \mid \mathbf{x}).$$

which describes our uncertainty about the parameter θ after observing the data. In other words, this distribution describes which parameter values remain plausible after combining the prior information with the observed data.

$$\text{MLE: } \mathbf{x} \longrightarrow \hat{\theta}_{\text{MLE}},$$
$$\text{MAP: } (\mathbf{x}, p(\theta)) \longrightarrow \hat{\theta}_{\text{MAP}},$$
$$\text{Bayesian inference: } (\mathbf{x}, p(\theta)) \longrightarrow p(\theta \mid \mathbf{x}).$$

The posterior distribution $p(\theta \mid \mathbf{x})$ contains more information than a single point estimate. For example, two datasets may produce similar estimates of θ but very different levels of uncertainty. A narrow posterior distribution indicates that the data constrain θ quite strongly, whereas a broad posterior indicates substantial remaining uncertainty.

Thus, Bayesian inference allows us to address not only

What value of θ is most plausible?

but also

How uncertain are we about θ?

This distinction will be important when we begin updating parameter distributions as additional data are observed.